

Exploring the Identity:

$$G^* = \sqrt{2\pi} \cdot \theta_3(e^{-\pi})^2$$

This is a mathematically beautiful identity that sits at the deep intersection of modular forms, elliptic integrals, continuous probability, and discrete geometry.

By unpacking the expression $G^* = \sqrt{2\pi} \cdot \theta_3(e^{-\pi})^2$ step-by-step, we can reveal its exact closed-form value, its elegant physical interpretations, and its direct relationship to **Gauss's constant**.

1. Evaluating the Jacobi Theta Function

The core of this identity is the Jacobi theta function $\theta_3(q)$, which is defined as an infinite sum of a Gaussian sequence over the integers:

$$\theta_3(q) = \sum_{n=-\infty}^{\infty} q^{n^2} \quad (1)$$

When we evaluate this at $q = e^{-\pi}$, we are looking at the highly symmetric “self-dual” point in the complex plane ($\tau = i$). Because the Gaussian curve is its own Fourier transform, evaluating it at this precise parameter yields a known analytical exact value in terms of the Euler Gamma function (Γ):

$$\theta_3(e^{-\pi}) = \frac{\pi^{1/4}}{\Gamma(3/4)} \quad (2)$$

To make this easier to work with, we can apply **Euler's reflection formula**, $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$. Setting $z = 1/4$:

$$\Gamma(1/4)\Gamma(3/4) = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2} \quad (3)$$

$$\implies \Gamma(3/4) = \frac{\pi\sqrt{2}}{\Gamma(1/4)} \quad (4)$$

Substituting this back into our evaluated theta function gives:

$$\theta_3(e^{-\pi}) = \frac{\pi^{1/4}\Gamma(1/4)}{\pi\sqrt{2}} = \frac{\Gamma(1/4)}{\sqrt{2}\pi^{3/4}} \quad (5)$$

2. Squaring and Multiplying by $\sqrt{2\pi}$

Next, the identity squares this theta value. Geometrically, squaring the 1D theta function gives us the sum of a Gaussian over a 2D square integer lattice ($\mathbb{Z}^2$):

$$\theta_3(e^{-\pi})^2 = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} e^{-\pi(m^2+n^2)} = \frac{\Gamma(1/4)^2}{2\pi^{3/2}} \quad (6)$$

Finally, we multiply by $\sqrt{2\pi}$ as defined in the identity to uncover the exact closed algebraic form of G^* :

$$G^* = \sqrt{2\pi} \cdot \frac{\Gamma(1/4)^2}{2\pi^{3/2}} \quad (7)$$

$$= \frac{\sqrt{2}\pi^{1/2}\Gamma(1/4)^2}{2\pi^{3/2}} \quad (8)$$

Simplifying the constants ($\pi^{1/2}/\pi^{3/2} = 1/\pi$) yields the true nature of the expression:

$$\boxed{G^* = \frac{\Gamma(1/4)^2}{\pi\sqrt{2}} \approx 2.9586751\dots} \quad (9)$$

3. The Bridge to Famous Historical Constants

Because G^* is driven by $\Gamma(1/4)^2$, it acts as a mathematical “hub” that algebraically connects several legendary 19th-century math constants:

- **Gauss’s Constant (G):** In 1799, Carl Friedrich Gauss discovered that the reciprocal of the arithmetic-geometric mean (AGM) of 1 and $\sqrt{2}$ yields a fundamental constant:

$$G = \frac{1}{\text{agm}(1, \sqrt{2})} = \frac{\Gamma(1/4)^2}{2\sqrt{2}\pi^{3/2}} \approx 0.8346$$

If we extract the Gamma values from our closed form, the identity reveals itself as a scaled version of Gauss’s constant:

$$\boxed{G^* = 2\sqrt{\pi} \cdot G} \quad (10)$$

- **The Lemniscate Constant (ϖ):** Just as π governs the perimeter of a circle, $\varpi \approx 2.622$ governs the perimeter of the Bernoulli Lemniscate (the ∞ -shaped curve). Because $\varpi = \pi G$, the constant scales directly with the arc length of the Lemniscate:

$$G^* = \frac{2}{\sqrt{\pi}} \cdot \varpi \quad (11)$$

- **The Beta Function:** The Beta function evaluated at one-quarter is $B(1/4, 1/4) = \frac{\Gamma(1/4)^2}{\sqrt{\pi}}$. This allows us to write the identity elegantly normalized by the square root of 2π :

$$G^* = \frac{1}{\sqrt{2\pi}} B(1/4, 1/4) \quad (12)$$

- **Complete Elliptic Integrals:** The complete elliptic integral of the first kind evaluated at $k = 1/\sqrt{2}$ yields $K(1/\sqrt{2}) = \frac{\Gamma(1/4)^2}{4\sqrt{\pi}}$. Thus:

$$G^* = \frac{2\sqrt{2}}{\sqrt{\pi}} K(1/\sqrt{2}) \quad (13)$$

4. The Continuous $\times$ Discrete Interpretation

Perhaps the most poetic way to look at this identity is as a direct collision of continuous probability and discrete lattice mathematics.

Look at the two distinct parts of the original formula:

1. $\sqrt{2\pi}$ is the exact total area/normalization factor of the **1D continuous standard Gaussian integral**: $\int_{-\infty}^{\infty} e^{-x^2/2} dx$.
2. $\theta_3(e^{-\pi})^2$ is the **2D discrete Gaussian partition function** acting over a square integer grid.

Therefore, the identity can conceptually be rewritten as:

$$G^* = \left(\int_{-\infty}^{\infty} e^{-x^2/2} dx \right) \times \left(\sum_{(m,n) \in \mathbb{Z}^2} e^{-\pi(m^2+n^2)} \right) \quad (14)$$

In Summary:

The identity miraculously takes the continuous volume of a normal probability distribution ($\sqrt{2\pi}$), multiplies it by the discrete heat kernel of a 2D integer lattice ($\theta_3(e^{-\pi})^2$), and neatly produces a fundamental constant $\left(\frac{\Gamma(1/4)^2}{\pi\sqrt{2}} \right)$ that is permanently tethered to Gauss's arithmetic-geometric mean.